角谷猜想最接近严格证明的测试（逆向三叉树） Version1.1.6
原创：豆角五谷 2026-11-01

①原文再续，书接上回，上回讲到，逆生长二叉树，自然数 M 飘流而下，有起点但找不到终点。现在尝试用逆生长三叉树，自然数 M 也是想飘流而下，发现有终点找不到起点。角谷猜想，从 2^n 倒过来推理，是 M=(2^n-1)/3 这样的数，才产生 3M+1=2^n，为终点最后一步的，这样的数形如 10101...阴阳相间的二进制数。现在，不停的逆行逆流而上，能否找到起点呢，这即是总思路，最简单的起步也。方向是不是正确的，也是未知的，科学就是这样探索的，长叹息以掩泪兮，吾将上下而求索也。
② M=(2^n-1)/3 这样的数，才产生 3M+1=2^n，为终点最后一步的，这样的数形如 10101...阴阳相间的二进制数。M 本身 mod3 时，余数可以为 0,1,2 三种，这三种都可能有的。
如果 M=3k，这时逆流，没有后果的，或称为这三叉树此枝 ID 已经结出果实了，逆流只能得到(2^n)*M 这样的偶数，不会出现奇数的。
如果 M=3k+2，这时逆流，乘二减一除三，产生为 P=((2^i)*M-1)/3，i 为奇数，这时，3P+1 后除以 2^i 为 M，M 或暂称为 K 核，P 或暂称为 K 簇。i 为任一奇数，数量为阿列夫零。
因为，P=((2^i)*M-1)/3=((2^i)*3k+(2^(i+1)-1))/3=（2^i)*k+(4^t-1)/3，因为 4^t=(3+1)^t，贾宪三角杨辉三角帕斯卡三角组合数知，(4^t-1)除三必为自然数。
如果 M=3k+1，这时逆流，乘四减一除三，产生为 Q=((2^j)*M-1)/3，j 为偶数，这时，3Q+1 后除以 2^j 为 M，M 或暂称为 K 核，P 或暂称为 K 簇。j 为任一偶数，数量为阿列夫零。
因为，Q=((2^j)*M-1)/3=((2^j)*3k+(2^j-1))/3=（2^j)*k+(4^t-1)/3，因为 4^t=(3+1)^t，贾宪三角杨辉三角帕斯卡三角组合数知，(4^t-1)除三必为自然数。
③上述② M 仅逆生长一次，不停地逆生长，就会产生三叉树。三叉树一样，象二叉树的枝条一样可以标记 ID，二叉树的枝条 ID 为二进制，此三叉树的枝条 ID 用三进制标记则十分自然也。
任一个三进制数 m，从左到右扫描，如果找到一个为 0，后面就不要扫描了，对应三叉树上一个 ID，此枝条 ID 的结点为 3 的倍数，后面没有奇数，只有偶数的了。
如果没有扫描出一个 0，则此三进制数 m 与三叉树必有一个 ID 相对应，必存在的，因为，这三叉树是无限逆生长的，且每个节点全部具备三分制 3k,3k+1,3k+2，与 n 分为 i 奇 j 偶有关，
((2^n)M-1)/3。这个可以用树形结构图来显示，TreeView 编程，暂不理，并不难，不想理，说明下而已。
④每一个 ID 有一个核心或暂称为 K 核，产生 K 簇，数量阿列夫零，但每一个 ID 又与自然数三进制 m 对应，总量仍然阿列夫零，此为《集合论》中基本定理，对角线定理，排列坐标轴，沿对角线标记 1,2,3,....一样的数量级，集合的势，阿列夫零也。
⑤此三叉树的 ID 对应所有三进制数，没有一个三进制数在三叉树上找不到对应 ID 的，所以，所有被三整除的数，必在三叉树上对应的 0 位所在的枝条上找到一个 ID 对应。
如果自然数 M 角谷运算无限，则在三叉树上找不到对应 ID。M 必可以逆行，M 或本身是 3M 点或逆行一步就可找到 3M 点，这个 3M 点必可三叉树上找到对应 ID，矛盾。所以不存在 M 可以无限长乎。此为好似是，好似不太是，留给读者看下如何判断？？？？

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角谷猜想最接近严格证明的测试 Version1.1.5
原创：吴素波 2026-10-24
①前文再续，书接上一回，上回讲到，角谷猜想从框架上架构上，应是正确方向，大约差不多证明，后来，忽又悟出，角谷逆向二叉树中，每次逆生长时，生出的阴阳二分法二分枝，对应一个二进制数 W，每节分支的 Wi 象是每节枝的标记 ID 一样，是一个二进制数，与分枝的深度递归的层次也有关，与二次归纳法投入的数 M 也有关，于是，悟得一个定理，称为：吴素波定理，用以纪念我伟大的母亲。
吴素波定理：在角谷逆向二叉树中，每次系数二分法出来的递归深度，每条分枝对应一个二进制数 ID，这个 ID 与二次数学归纳法投入验证的自然数 M 之间的关系就是，当 ID>M 时，二叉树对 M 的处理已完成，绝不会出现在 ID>M 的分枝中。这个我称为角谷猜想的核心映射原理。简称素波定理。或简称吴川定理，吴定理，Wu 定理。
==在与百度文心助手 AI 的网页版聊天中，得知测试 27 等数，不对了，超出二进制头部，仍然没有处理完，而且特别超长待机的那种。于是，又开始了抑郁式思考了。
②忽然想到，在 27 的前面不停补 0，0 头，零头，是不是可以呢，这显然补个 0 头，要改代码的，测试，居然成了，忽又想到，这与二叉树对应可能不是很完美，因为，如果补集，前面专门补 1，是不是也一样，测试表明，居然也是一样

的，再测，前面 0 或 1 是随机的，竟然也是一样的，这讲明，树与子树是相似的，吴定理中，二进制扫描未到头部已经结束的，与现在补上前缀的，就是映射相似的全树与子树乎。于是，称之为“广义素波定理”或广义吴定理广义 Wu 定理。修改代码如下：

③
```
procedure TfrmMain.btnRUN1Click(Sender: TObject);
var
  www: integer;
  varv: string;
begin
  self.Memo1.Lines.Clear;
  self.linelist := TStringList.Create;
  rate := 0.0;
  varv := '100111000111011';
  if EditM.Text = '' then
    if inputquery('请输入要处理的自然数 M：',
      '处理全路径将显示：', varv) then
      EditM.Text := IntToStr(getIntFromByteString(varv));
  EditM2.Text := varv;
   Randomize; // 只准用一次，在递归函数外部调用，内部不可乎
  www := GreenBeanGrain2(StrToInt(EditM.Text), '1', 1, 2, 1, 1, 2, 1, 1);
  begin
    ShowMessage('ok=' + IntToStr(www));
    application.ProcessMessages;
    self.Memo1.Lines.Assign(self.linelist);
    ShowMessage('Memo=ok=' + IntToStr(linelist.Count));
    application.ProcessMessages;
    self.Memo1.Lines.SaveToFile('GreenBeanGrain.txt');
    ShowMessage('txt=ok');
  end;
  linelist.Free;
end;
=====
function TfrmMain.GreenBeanGrain2(M: int64; ID: string;
  deep, aisGreen, xisBean, bisGrain: int64; const Green, Bean, Grain: int64): integer;
var
  // a, b, x: int64;
  a, b, x, mm: int64;
  //a, b, x: integer;
  gbg: integer;
  aa: integer;
  one, head: string;
begin
  gbg := 0;
  deep := deep * 2;
  a := aisGreen;
  x := xisBean;
  b := bisGrain;
  aa := 0;
  if gbg = 0 then
  begin
    while (((a mod 2) = 0) and ((b mod 2) = 0)) do
    begin
```

```
    if (((a mod 2) = 0) and ((b mod 2) = 0)) then
    begin
      if (((a mod 3) = 0) and ((b mod 3) = 1) and (a >= 3) and (b > 3)) then
      begin
        a := a div 3;
        b := (b - 1) div 3;
        aa := aa + 1;

      end;
      if a < Green then
        gbg := 1
      else if ((a = Green) and (b < Grain)) then
        gbg := 1;
    end;
    if (((a mod 2) = 0) and ((b mod 2) = 0)) then
    begin
      a := a div 2;
      b := b div 2;
    end;
  end;
  if a < Green then
    gbg := 1
  else if ((a = Green) and (b < Grain)) then
    // gbg := 1;
    gbg := 1;
  ////else if (b < Grain) then
  ///    gbg := 1;  //此处断点，一个也没有
end;

 if aa = 0 then
  if gbg = 0 then
  begin
    while (((a mod 2) = 0) and ((b mod 2) = 1)) do
    begin
      a := a * 3;
      b := b * 3 + 1;

      while (((a mod 2) = 0) and ((b mod 2) = 0)) do
      begin
        a := a div 2;
        b := b div 2;
        if (((a mod 2) = 0) and ((b mod 2) = 0)) then
        begin
          if (((a mod 3) = 0) and ((b mod 3) = 1) and (a >= 3) and (b > 3)) then
          begin
            a := a div 3;
            b := (b - 1) div 3;
            aa := aa + 1;

          end;
          if a < Green then
```

```
          gbg := 1
        else if ((a = Green) and (b < Grain)) then
          gbg := 1;
      end;

    end;
    if aa > 0 then break;
  end;
  if a < Green then
    gbg := 1
  else if ((a = Green) and (b < Grain)) then
    gbg := 1;
end;



// if 1=0 then
if gbg = 1 then
begin
  //if not ((a > 0) and (b>0) and  (Green > 0) and (Grain>0)) then
  if ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
    //if not ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
  begin
    // rate:=rate+(1.0/deep);
    // rate := rate + (2.0 / deep);
     one := inttostr(linelist.Count) +'行:  '+ IntToStr(Green) + ' * N' + IntToStr(Bean) + ' + ' +
      IntToStr(Grain) + ' -->  ' + IntToStr(a) + ' * M' + IntToStr(x) +
      ' + ' + IntToStr(b) + ' ok  branID=' + ID + '=' + IntToStr(
      getintFromByteString(ID)) + ' M=' + IntToStr(M) + '=' +
      getByteStringFromint(M);//+ '  rate=' + floattostr(rate);
    if getintFromByteString(ID) > M then
      one := one + '   ID=M+' + IntToStr(getintFromByteString(ID) - M);
    linelist.add(one);
    //self.Memo1.Lines.add(one);
    //application.ProcessMessages;
    //  if linelist.Count>290980 then
    //     application.ProcessMessages;
  end;
end;
//if getintFromByteString(ID) <= M then
if gbg = 0 then
  //if not ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
  if ((a > 0) and (b > 0) and (Green > 0) and (Grain > 0)) then
  begin
    // if deep <= trunc(power(2, 49)) then
    begin
      //  gbg := GreenBeanGrain(aisGreen * 2, xisBean + 1, bisGrain, Green *
      //   2, Bean + 1, Grain);
      //rate := rate + (2.0 / deep);
       one := inttostr(linelist.Count) +'行:  '+ IntToStr(Green) + ' * N' + IntToStr(Bean) + ' + ' +
```

```
  IntToStr(Grain) + ' -->  ' + IntToStr(a) + ' * M' + IntToStr(x) +
  ' + ' + IntToStr(b) + '   branID=' + ID + '=' + IntToStr(
  getintFromByteString(ID)) + ' M=' + IntToStr(M) + '=' +
  getByteStringFromint(M);// + '  rate=' + floattostr(rate);
if getintFromByteString(ID) > M then
  one := one + '   ID=M+' + IntToStr(getintFromByteString(ID) - M);
linelist.add(one);
if getintFromByteString(ID) < M then
begin
  if rightstr(getByteStringFromInt(M), length(ID) + 1) = '0' + ID then
    gbg := GreenBeanGrain2(M, '0' + ID, deep, a * 2, x + 1, b,
      Green * 2, Bean + 1, Grain);
  //  if  (aisGreen * 2 mod 3)=0 then
  //     application.ProcessMessages;
  //  gbg := gbg * GreenBeanGrain(aisGreen * 2, xisBean + 1, aisGreen +
  //    bisGrain, Green * 2, Bean + 1, Green + Grain);

  if rightstr(getByteStringFromInt(M), length(ID) + 1) = '1' + ID then
    gbg := gbg * GreenBeanGrain2(M, '1' + ID, deep, a * 2, x +
      1, a + b, Green * 2, Bean + 1, Green + Grain);
end;
//Randomize;
// 关键：初始化随机种子，确保每次运行结果不同
head := inttostr(Random(2)); // 生成 0 或 1
//head := '0'; // 0 或 1
if (length(ID) = length(getByteStringFromInt(M))) then
  if getintFromByteString(ID) = M then
  begin
    if head='1' then
    M:=M+trunc(power(2,length(ID)-1));
    gbg := GreenBeanGrain2(M, head + ID, deep, a * 2, x + 1,
      b, Green * 2, Bean + 1, Grain);
    //  if  (aisGreen * 2 mod 3)=0 then
    //     application.ProcessMessages;
    //  gbg := gbg * GreenBeanGrain(aisGreen * 2, xisBean + 1, aisGreen +
    //    bisGrain, Green * 2, Bean + 1, Green + Grain);


    // gbg := gbg * GreenBeanGrain2(M, '1' + ID, deep, a * 2, x +
    //   1, a + b, Green * 2, Bean + 1, Green + Grain);
  end;
if ID = '011011' then
begin
  application.ProcessMessages;
  // showmessage(inttostr(length(ID) )+'   '+inttostr(length(getByteStringFromInt(M))));

end;
//Randomize;
// 关键：初始化随机种子，确保每次运行结果不同
head := inttostr(Random(2)); // 生成 0 或 1
//head := '0'; // 0 或 1
```

```
      //if  ((length(ID) > length(getByteStringFromInt(M))) and (length(ID) < (length(getByteStringFromInt(M))+250) )) then
      if (length(ID) > length(getByteStringFromInt(M))) then
      begin
        if head='1' then
          M:=M+trunc(power(2,length(ID)-1));
        gbg := GreenBeanGrain2(M, head + ID, deep, a * 2, x + 1, b,
          Green * 2, Bean + 1, Grain);
      //if gbg = 0 then
      //  ShowMessage('error');
      //if gbg = 0 then
      //  application.ProcessMessages;
      end;

    end;
    if deep = trunc(power(2, 50)) then
    begin
      //gbg := 1;
      // gbg := 0;
    end;
  end;

  Result := gbg;
end; gbg;
end;
```

④将 27 代入，运算结果如下：共 59 步：

```
2 * N1 + 1 -->  3 * M1 + 2   branID=1=1 M=27=11011
4 * N2 + 3 -->  9 * M2 + 8   branID=11=3 M=27=11011
8 * N3 + 3 -->  9 * M3 + 4   branID=011=3 M=27=11011
16 * N4 + 11 -->  27 * M4 + 20   branID=1011=11 M=27=11011
32 * N5 + 27 -->  81 * M5 + 71   branID=11011=27 M=27=11011
64 * N6 + 27 -->  243 * M6 + 107   branID=011011=27 M=27=11011
128 * N7 + 27 -->  729 * M7 + 161   branID=0011011=27 M=27=11011
256 * N8 + 27 -->  2187 * M8 + 242   branID=00011011=27 M=27=11011
512 * N9 + 27 -->  2187 * M9 + 121   branID=000011011=27 M=27=11011
1024 * N10 + 27 -->  6561 * M10 + 182   branID=0000011011=27 M=27=11011
2048 * N11 + 27 -->  6561 * M11 + 91   branID=10000011011=1051 M=539=1000011011   ID=M+512
4096 * N12 + 27 -->  19683 * M12 + 137   branID=110000011011=3099 M=1563=11000011011   ID=M+1536
8192 * N13 + 27 -->  59049 * M13 + 206   branID=0110000011011=3099 M=1563=11000011011   ID=M+1536
16384 * N14 + 27 -->  59049 * M14 + 103   branID=10110000011011=11291 M=5659=1011000011011   ID=M+5632
32768 * N15 + 27 -->  177147 * M15 + 155   branID=010110000011011=11291 M=5659=1011000011011   ID=M+5632
65536 * N16 + 27 -->  531441 * M16 + 233   branID=0010110000011011=11291 M=5659=1011000011011   ID=M+5632
131072 * N17 + 27 -->  1594323 * M17 + 350   branID=10010110000011011=76827 M=38427=1001011000011011
ID=M+38400
262144 * N18 + 27 -->  1594323 * M18 + 175   branID=010010110000011011=76827 M=38427=1001011000011011
ID=M+38400
524288 * N19 + 27 -->  4782969 * M19 + 263   branID=0010010110000011011=76827 M=38427=1001011000011011
ID=M+38400
1048576 * N20 + 27 -->  14348907 * M20 + 395   branID=00010010110000011011=76827 M=38427=1001011000011011
ID=M+38400
2097152 * N21 + 27 -->  43046721 * M21 + 593   branID=000010010110000011011=76827 M=38427=1001011000011011
ID=M+38400
4194304 * N22 + 27 -->  129140163 * M22 + 890   branID=1000010010110000011011=2173979
M=1087003=100001001011000011011   ID=M+1086976
8388608 * N23 + 27 -->  129140163 * M23 + 445   branID=11000010010110000011011=6368283
M=3184155=110001001011000011011   ID=M+3184128
```

16777216 * N24 + 27 --> 387420489 * M24 + 668 branID=011000010010110000011011=6368283
M=3184155=110001001011000011011 ID=M+3184128
33554432 * N25 + 27 --> 387420489 * M25 + 334 branID=101100001001011000011011=23145499
M=11572763=101100001001011000011011 ID=M+11572736
67108864 * N26 + 27 --> 258280326 * M26 + 111 branID=010110000100101100011011=23145499
M=11572763=101100001001011000011011 ID=M+11572736
134217728 * N27 + 27 --> 1162261467 * M27 + 251 branID=001011000010010110000011011=23145499
M=11572763=101100001001011000011011 ID=M+11572736
268435456 * N28 + 27 --> 3486784401 * M28 + 377 branID=100101100001001011000011011=157363227
M=78681627=100101100001001011000011011 ID=M+78681600
536870912 * N29 + 27 --> 10460353203 * M29 + 566 branID=010010110000100101100011011=157363227
M=78681627=100101100001001011000011011 ID=M+78681600
1073741824 * N30 + 27 --> 10460353203 * M30 + 283 branID=101001011000010010110000011011=694234139
M=347117083=101001011000010010110000011011 ID=M+347117056
2147483648 * N31 + 27 --> 31381059609 * M31 + 425 branID=010100101100001001011000011011=694234139
M=347117083=101001011000010010110000011011 ID=M+347117056
4294967296 * N32 + 27 --> 94143178827 * M32 + 638 branID=001010010110000100101100011011=694234139
M=347117083=101001011000010010110000011011 ID=M+347117056
8589934592 * N33 + 27 --> 94143178827 * M33 + 319 branID=000101001011000010010110000011011=694234139
M=347117083=101001011000010010110000011011 ID=M+347117056
17179869184 * N34 + 27 --> 282429536481 * M34 + 479 branID=100010100101100001001011000011011=9284168731
M=4642084379=1000101001011000010010110000011011 ID=M+4642084352
34359738368 * N35 + 27 --> 847288609443 * M35 + 719 branID=010001010010110000100101100011011=9284168731
M=4642084379=1000101001011000010010110000011011 ID=M+4642084352
68719476736 * N36 + 27 --> 2541865828329 * M36 + 1079
branID=001000101001011000010010110000011011=9284168731 M=4642084379=1000101001011000010010110000011011
ID=M+4642084352
137438953472 * N37 + 27 --> 7625597484987 * M37 + 1619
branID=100100010100101100001001011000011011=78003645467
M=39001822747=1001000101001011000010010110000011011 ID=M+39001822720
274877906944 * N38 + 27 --> 22876792454961 * M38 + 2429
branID=010010001010010110000100101100011011=78003645467
M=39001822747=1001000101001011000010010110000011011 ID=M+39001822720
549755813888 * N39 + 27 --> 68630377364883 * M39 + 3644
branID=101001000101001011000010010110000011011=352881552411
M=176440776219=1010010001010010110000100101100011011 ID=M+176440776192
1099511627776 * N40 + 27 --> 68630377364883 * M40 + 1822
branID=010100100010100101100001001011000011011=352881552411
M=176440776219=1010010001010010110000100101100011011 ID=M+176440776192
2199023255552 * N41 + 27 --> 45753584909922 * M41 + 607
branID=101010010001010010110000100101100011011=1452393180187
M=726196590107=1010100100010100101100010010110000011011 ID=M+726196590080
4398046511104 * N42 + 27 --> 205891132094649 * M42 + 1367
branID=110101001000101001011000010010110000011011=3651416435739
M=1825708217883=1101010010001010010110000100101100011011 ID=M+1825708217856
8796093022208 * N43 + 27 --> 617673396283947 * M43 + 2051
branID=011010100100010100101100001001011000011011=3651416435739
M=1825708217883=1101010010001010010110000100101100011011 ID=M+1825708217856
17592186044416 * N44 + 27 --> 1853020188851841 * M44 + 3077
branID=001101010010001010010110000100101100011011=3651416435739
M=1825708217883=1101010010001010010110000100101100011011 ID=M+1825708217856
35184372088832 * N45 + 27 --> 5559060566555523 * M45 + 4616
branID=000110101001000101001011000010010110000011011=3651416435739
M=1825708217883=1101010010001010010110000100101100011011 ID=M+1825708217856
70368744177664 * N46 + 27 --> 5559060566555523 * M46 + 2308
branID=000011010100100010100101100001001011000011011=3651416435739
M=1825708217883=1101010010001010010110000100101100011011 ID=M+1825708217856
140737488355328 * N47 + 27 --> 3706040377703682 * M47 + 769
branID=100001101010010001010010110000100101100011011=74020160613403
M=37010080306715=1000011010100100010100101100001001011000011011 ID=M+37010080306688

281474976710656 * N48 + 27 --> 5559060566555523 * M48 + 577
branID=0100001101010010001010010110000100101100000011011=74020160613403
M=37010080306715=10000110101001000101001011000010010110000011011 ID=M+37010080306688
562949953421312 * N49 + 27 --> 16677181699666569 * M49 + 866
branID=1010000110101001000101001011000010010110000011011=355495137324059
M=177747568662043=10100001101010010001010010110000100101100000011011 ID=M+177747568662016
1125899906842624 * N50 + 27 --> 16677181699666569 * M50 + 433
branID=0101000011010100100010100101100001001011000011011=355495137324059
M=177747568662043=10100001101010010001010010110000100101100000011011 ID=M+177747568662016
2251799813685248 * N51 + 27 --> 50031545098999707 * M51 + 650
branID=0010100001101010010001010010110000100101100000011011=355495137324059
M=177747568662043=10100001101010010001010010110000100101100000011011 ID=M+177747568662016
4503599627370496 * N52 + 27 --> 50031545098999707 * M52 + 325
branID=1001010000110101001000101001011000010010110000011011=2607294951009307
M=1303647475504667=10010100001101010010001010010110000100101100000011011 ID=M+1303647475504640
9007199254740992 * N53 + 27 --> 150094635296999121 * M53 + 488
branID=0100101000011010100100010100101100001001011000011011=2607294951009307
M=1303647475504667=10010100001101010010001010010110000100101100000011011 ID=M+1303647475504640
18014398509481984 * N54 + 27 --> 150094635296999121 * M54 + 244
branID=0010010100001101010010001010010110000100101100000011011=2607294951009307
M=1303647475504667=10010100001101010010001010010110000100101100000011011 ID=M+1303647475504640
36028797018963968 * N55 + 27 --> 100063090197999414 * M55 + 81
branID=1001001010000110101001000101001011000010010110000011011=20621693460491291
M=10310846730245659=10010010100001101010010001010010110000100101100000011011 ID=M+10310846730245632
72057594037927936 * N56 + 27 --> 150094635296999121 * M56 + 61
branID=1100100101000011010100100010100101100001001011000011011=56650490479455259
M=28325245239727643=11001001010000110101001000101001011000010010110000011011 ID=M+28325245239727616
144115188075855872 * N57 + 27 --> 450283905890997363 * M57 + 92
branID=1110010010100001101010010001010010110000100101100000011011=128708084517383195
M=64354042258691611=11100100101000011010100100010100101100001001011000011011 ID=M+64354042258691584
288230376151711744 * N58 + 27 --> 450283905890997363 * M58 + 46
branID=0111001001010000110101001000101001011000010010110000011011=128708084517383195
M=64354042258691611=11100100101000011010100100010100101100001001011000011011 ID=M+64354042258691584
576460752303423488 * N59 + 27 --> 300189270593998242 * M59 + 15 ok
branID=1011100100101000011010100100010100101100001001011000011011=416938460669094939
M=208469230334547483=10111001001010000110101001000101001011000010010110000011011
ID=M+208469230334547456

如果将随机前缀改成每次只补'0',

```
//Randomize;
// 关键：初始化随机种子，确保每次运行结果不同
//head := inttostr(Random(2)); // 生成 0 或 1
head := '0'; // 0 或 1
```

则结果如下不变，一样 59 步：
2 * N1 + 1 --> 3 * M1 + 2 branID=1=1 M=27=11011
4 * N2 + 3 --> 9 * M2 + 8 branID=11=3 M=27=11011
8 * N3 + 3 --> 9 * M3 + 4 branID=011=3 M=27=11011
16 * N4 + 11 --> 27 * M4 + 20 branID=1011=11 M=27=11011
32 * N5 + 27 --> 81 * M5 + 71 branID=11011=27 M=27=11011
64 * N6 + 27 --> 243 * M6 + 107 branID=011011=27 M=27=11011
128 * N7 + 27 --> 729 * M7 + 161 branID=0011011=27 M=27=11011
256 * N8 + 27 --> 2187 * M8 + 242 branID=00011011=27 M=27=11011
512 * N9 + 27 --> 2187 * M9 + 121 branID=000011011=27 M=27=11011
1024 * N10 + 27 --> 6561 * M10 + 182 branID=0000011011=27 M=27=11011
2048 * N11 + 27 --> 6561 * M11 + 91 branID=00000011011=27 M=27=11011
4096 * N12 + 27 --> 19683 * M12 + 137 branID=000000011011=27 M=27=11011
8192 * N13 + 27 --> 59049 * M13 + 206 branID=0000000011011=27 M=27=11011
16384 * N14 + 27 --> 59049 * M14 + 103 branID=00000000011011=27 M=27=11011
32768 * N15 + 27 --> 177147 * M15 + 155 branID=000000000011011=27 M=27=11011
65536 * N16 + 27 --> 531441 * M16 + 233 branID=0000000000011011=27 M=27=11011
131072 * N17 + 27 --> 1594323 * M17 + 350 branID=00000000000011011=27 M=27=11011

262144 * N18 + 27 --> 1594323 * M18 + 175 branID=0000000000000011011=27 M=27=11011
524288 * N19 + 27 --> 4782969 * M19 + 263 branID=0000000000000011011=27 M=27=11011
1048576 * N20 + 27 --> 14348907 * M20 + 395 branID=0000000000000011011=27 M=27=11011
2097152 * N21 + 27 --> 43046721 * M21 + 593 branID=0000000000000011011=27 M=27=11011
4194304 * N22 + 27 --> 129140163 * M22 + 890 branID=0000000000000011011=27 M=27=11011
8388608 * N23 + 27 --> 129140163 * M23 + 445 branID=0000000000000011011=27 M=27=11011
16777216 * N24 + 27 --> 387420489 * M24 + 668 branID=0000000000000011011=27 M=27=11011
33554432 * N25 + 27 --> 387420489 * M25 + 334 branID=0000000000000011011=27 M=27=11011
67108864 * N26 + 27 --> 258280326 * M26 + 111 branID=0000000000000011011=27 M=27=11011
134217728 * N27 + 27 --> 1162261467 * M27 + 251 branID=0000000000000011011=27 M=27=11011
268435456 * N28 + 27 --> 3486784401 * M28 + 377 branID=0000000000000011011=27 M=27=11011
536870912 * N29 + 27 --> 10460353203 * M29 + 566 branID=0000000000000011011=27 M=27=11011
1073741824 * N30 + 27 --> 10460353203 * M30 + 283 branID=0000000000000011011=27 M=27=11011
2147483648 * N31 + 27 --> 31381059609 * M31 + 425 branID=0000000000000011011=27 M=27=11011
4294967296 * N32 + 27 --> 94143178827 * M32 + 638 branID=0000000000000011011=27 M=27=11011
8589934592 * N33 + 27 --> 94143178827 * M33 + 319 branID=0000000000000011011=27 M=27=11011
17179869184 * N34 + 27 --> 282429536481 * M34 + 479 branID=0000000000000011011=27
M=27=11011
34359738368 * N35 + 27 --> 847288609443 * M35 + 719 branID=0000000000000011011=27
M=27=11011
68719476736 * N36 + 27 --> 2541865828329 * M36 + 1079 branID=0000000000000011011=27
M=27=11011
137438953472 * N37 + 27 --> 7625597484987 * M37 + 1619 branID=0000000000000011011=27
M=27=11011
274877906944 * N38 + 27 --> 22876792454961 * M38 + 2429 branID=0000000000000011011=27
M=27=11011
549755813888 * N39 + 27 --> 68630377364883 * M39 + 3644 branID=0000000000000011011=27
M=27=11011
1099511627776 * N40 + 27 --> 68630377364883 * M40 + 1822 branID=0000000000000011011=27
M=27=11011
2199023255552 * N41 + 27 --> 45753584909922 * M41 + 607
branID=0000000000000011011=27 M=27=11011
4398046511104 * N42 + 27 --> 205891132094649 * M42 + 1367
branID=0000000000000011011=27 M=27=11011
8796093022208 * N43 + 27 --> 617673396283947 * M43 + 2051
branID=0000000000000011011=27 M=27=11011
17592186044416 * N44 + 27 --> 1853020188851841 * M44 + 3077
branID=0000000000000011011=27 M=27=11011
35184372088832 * N45 + 27 --> 5559060566555523 * M45 + 4616
branID=0000000000000011011=27 M=27=11011
70368744177664 * N46 + 27 --> 5559060566555523 * M46 + 2308
branID=0000000000000011011=27 M=27=11011
140737488355328 * N47 + 27 --> 3706040377703682 * M47 + 769
branID=0000000000000011011=27 M=27=11011
281474976710656 * N48 + 27 --> 5559060566555523 * M48 + 577
branID=0000000000000011011=27 M=27=11011
562949953421312 * N49 + 27 --> 16677181699666569 * M49 + 866
branID=0000000000000011011=27 M=27=11011
1125899906842624 * N50 + 27 --> 16677181699666569 * M50 + 433
branID=0000000000000011011=27 M=27=11011
2251799813685248 * N51 + 27 --> 50031545098999707 * M51 + 650
branID=0000000000000011011=27 M=27=11011
4503599627370496 * N52 + 27 --> 50031545098999707 * M52 + 325
branID=0000000000000011011=27 M=27=11011
9007199254740992 * N53 + 27 --> 150094635296999121 * M53 + 488
branID=0000000000000011011=27 M=27=11011
18014398509481984 * N54 + 27 --> 150094635296999121 * M54 + 244
branID=0000000000000011011=27 M=27=11011
36028797018963968 * N55 + 27 --> 100063090197999414 * M55 + 81
branID=0000000000000011011=27 M=27=11011
72057594037927936 * N56 + 27 --> 150094635296999121 * M56 + 61
branID=0000000000000011011=27 M=27=11011

144115188075855872 * N57 + 27 --> 450283905890997363 * M57 + 92
branID=00000000000000000000000000000000000000000000000000000011011=27 M=27=11011
288230376151711744 * N58 + 27 --> 450283905890997363 * M58 + 46
branID=00000000000000000000000000000000000000000000000000000011011=27 M=27=11011
576460752303423488 * N59 + 27 --> 300189270593998242 * M59 + 15 ok
branID=00000000000000000000000000000000000000000000000000000011011=27 M=27=11011
⑤发现，在 27 的前头，补前缀，随机的，每次随机 0 或 1，不影响结果，但是，如果一次性固定的定值定数前缀加到 27 前，就会变化改变了。所以不可以有定数。要每次随机仅一位，不影响结果。
M 自然数不参与运算，只参与对比与飘流时分支选择，M 变不变，二叉树是虚树都是不会变化的。
⑥这与二叉树的流程一一对应，当自然数 M 在进入树洞前，在树洞中，无论随机飘流到哪个分枝，必定有一个挂住落脚点乎。如果一开始就前缀加个定数定值，在进入树洞前就已改变了自然数 M 的质量数 M 了乎，所以如此解释，象质量自然数 M 的小球，在进入树洞中随机随分支飘流乎。
⑦如此好似解释了角谷猜想二叉树，好似仍不太踏实，暂放一边，这或对应一颗树而已，3n+1 如果扩展成 5n+1。。。(an+b)/c 则为树林乎，
这个更难了要对比比较乎。如果不同树种呢，就象任意多项式 F(n)或无穷级数时，这树种长成啥样呢，不知道了，不可知论乎。
⑧所以，仍然臆想式自称：本世纪最伟大的数学家之一乎。